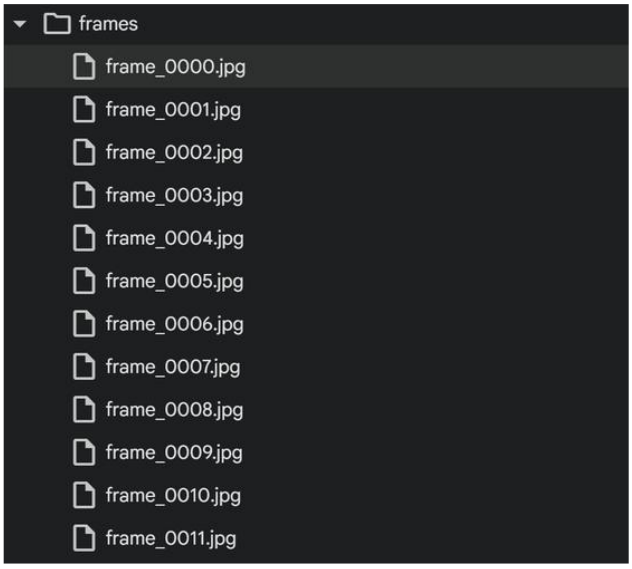
1. **Frame Extraction Output**

   Stage: Frame Extraction (OpenCV)

- Video is split into individual frames
- Each frame is processed independently



2. **Detection Output**

   Stage: YOLOv8 Detection

- Bounding boxes are drawn around license plates
- Model identifies plate location with confidence

3. **Pre-processed Plate Image**

   Stage: Image Preprocessing

- Cropped plate is converted to grayscale
- Thresholding improves text clarity
- Noise is reduced for better OCR accuracy
- Output includes Grayscale / thresholded image

4. **OCR Output**

   Stage: Text Recognition (Tesseract)

- Text is extracted from processed plate image
- Output includes: License plate number

```
Detected: MH20DV2366
```